TCS NQT QUANTITATIVE STRATEGY SERIES

Quantitative Aptitude: Number System Masterclass

Divisibility Rules, Numbers & Decimal Fractions, HCF & LCM Blueprint

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Section 1: Core Conceptual Clearance & Algebraic Shortcuts

The Number System forms the core foundation of the TCS NQT curriculum. Advanced sections frequently bypass direct questions to test combined properties—such as interlocking divisibility conditions or multi-stage remainder cycles. Relying on hit-and-trial methods will slow you down. The frameworks below are optimized for quick calculation under time constraints.

1. High-Yield Combined Divisibility Rules

TCS questions love to evaluate large numbers containing unknown placeholder variables (x, y) against composite divisors like 72, 88, or 99.

Divisibility Breakdown Matrix:

- **Divisibility by 8:** The last three digits of the number must form an integer perfectly divisible by 8. (Shortcut: If the hundreds digit is even, look only at the last 2 digits. If the hundreds digit is odd, add 4 to the last 2 digits and check).
- **Divisibility by 9:** The sum of all digits in the number must be a multiple of 9.
- **Divisibility by 11:** The absolute variance between the sum of digits at odd positions and the sum of digits at even positions must be either 0 or a multiple of 11.
- **Composite Rules:** To check for 72, test both 8 and 9 simultaneously (8×9). For 88, check 8 and 11 simultaneously (8×11).

2. Successive Division & Remainder Accumulation

When an unknown dividend is divided successively by a sequence of divisors (d_1, d_2, d_3) producing individual remainders (r_1, r_2, r_3), the final quotient from one step serves as the starting dividend for the next step.

The Absolute Net Remainder Formula:

To compute the direct total remainder when the original number is divided by the product divisor ($d_1 \times d_2 \times d_3$) all at once, use the synthetic chain equation:

$$\text{Net Remainder} = r_1 + (d_1 \times r_2) + (d_1 \times d_2 \times r_3)$$

3. Advanced HCF & LCM Core Theorems

Beyond the simple relationship $HCF \times LCM = \text{Product of Two Numbers}$, you must master the linear factor structure of co-prime sets.

The Co-Prime Factor Equation:

If two numbers have an HCF of H, they can be structured as $H \cdot a$ and $H \cdot b$, where a and b are strictly co-prime numbers ($HCF(a, b) = 1$).
• $LCM = H \times a \times b$
• **Bezout's Identity:** The HCF of any two integers x and y can always be represented as a linear combination: $HCF(x, y) = m \cdot x + n \cdot y$, where m and n are positive or negative integers.

4. Pure & Mixed Recurring Decimal Fractions

To convert recurring numbers (vinculum notations) into regular fractions efficiently without long algebraic setups, apply this direct rule:

$$0.ab\overline{cd} = \frac{\text{Complete Number (abcd)} - \text{Non-recurring Part (ab)}}{\text{Write '9' as many times as recurring digits} \cdot \text{Write '0' as many times as non-recurring decimal digits}}$$

Example: $0.3\overline{42} = \frac{342 - 3}{990} = \frac{339}{990}$.

Section 2: The Examiner's Trap Door - Where TCS Catches You

As assessment engineers, we purposefully design options that look correct but are based on common conceptual errors. Watch out for these four main traps in Number System questions.

TRAP 1: CONFUSING SUCCESSIVE VS. SIMULTANEOUS DIVISION

Consider the phrasing: *"A number when divided successively by 7, 11, and 13..."* versus *"A number divided by 7, 11, and 13 leaves remainders..."*. Simultaneous division requires finding the LCM of the divisors. Successive division means tracking the quotient step-by-step. If you confuse the two, you'll end up with an incorrect answer that matches Option A or B perfectly.

TRAP 2: THE BEZOUT COEFFICIENT SIGN FLIP

When matching an HCF to the linear expression $180m + 432n = HCF$, students often compute positive scalar values using the standard Extended Euclidean algorithm but forget that one coefficient **must be negative** to balance out a small positive result. A simple sign error here can lead you straight into a carefully placed trap option.

TRAP 3: SQUARE ROOTS OF FLOATING-POINT DECIMAL FRACTIONS

When evaluating expressions like $\sqrt{0.0001}$, students often count the zeros quickly and guess 0.001. **Correction:** Always convert the decimal to a fraction first: $\sqrt{1 / 10000} = 1 / 100 = 0.01$. Similarly, when adding decimals like $0.000314 + 0.000198 = 0.000512$, its cube root is 0.08 (since $0.08^3 = 0.000512$), not 0.02 or 0.8.

TRAP 4: CO-PRIME FACTOR PAIR LIMITS

When a problem provides the HCF and LCM and asks you to find the number of possible pairs, students often divide the LCM by the HCF to get the product $a \cdot b = K$, and then list all factors of K. This is a trap! The factor pairs (a, b) **must be strictly co-prime**. If you include a pair that shares a common factor, your count will be wrong.

Section 3: 10 High-Priority Questions & Comprehensive Solutions

Question 1: Composite Interlocking Variable Divisibility (Advanced Pattern)

Find the value of the expression $(4x - 3y)$ if a 9-digit corporate asset index number given by $785x3678y$ is perfectly divisible by 72.

A) 12

B) 10

C) 14

D) 8

SOLUTION EXPLANATION:

Since the number is divisible by 72, it must be divisible by both 8 and 9 simultaneously ($72 = 8 \times 9$).

• **Test for 8:** The last 3 digits, $78y$, must be divisible by 8.

Dividing 780 by 8: $780 = 97 \times 8 + 4$. To make it perfectly divisible, y must be 4 (since $784 / 8 = 98$). Thus, $y = 4$.

• **Test for 9:** The sum of all digits must be a multiple of 9.

$ext\{Sum\} = 7 + 8 + 5 + x + 3 + 6 + 7 + 8 + 4 = 48 + x$.

The next multiple of 9 after 48 is 54. Therefore, $48 + x = 54 \rightarrow x = 6$.

Substitute $x = 6$ and $y = 4$ into our target expression:

$4x - 3y = 4(6) - 3(4) = 24 - 12 = 12$.

Correct Answer: A

Question 2: Successive Division Remainder Convergence [cite: 2375]

An integer processing index is divided by 1001 $(= 7 \times 11 \times 13)$ by dividing it in succession by the individual prime factors 7, 11, and 13 [cite: 2375]. This process yields step-remainders of 4, 6, and 12 respectively [cite: 2375]. What would have been the single direct remainder if the original number was divided directly by 1001 [cite: 2376]?

A) 828 [cite: 2377]

B) 970 [cite: 2378]

C) 982 [cite: 2379]

D) 764 [cite: 2380]

SOLUTION EXPLANATION:

This is a direct application of Trap 1 and the synthetic chain remainder theorem.

Given divisors: $d_1 = 7, d_2 = 11, d_3 = 13$ [cite: 2375].

Given remainders: $r_1 = 4, r_2 = 6, r_3 = 12$ [cite: 2375].

Apply our net remainder formula:

$Net\ Remainder = r_1 + (d_1 \times r_2) + (d_1 \times d_2 \times r_3)$

Substitute the parameters systematically:

$Net\ Remainder = 4 + (7 \times 6) + (7 \times 11 \times 12)$

$Net\ Remainder = 4 + 42 + (77 \times 12)$

$Net\ Remainder = 46 + 924 = 970$ [cite: 2378].

The direct division remainder under the unified composite baseline is exactly 970 [cite: 2378].

Correct Answer: B [cite: 2378]

Question 3: Co-Prime Factor Multipliers [cite: 2310]

The Highest Common Factor (HCF) of two integer parameters is verified to be 17, and their Least Common Multiple (LCM) is recorded as 1309 [cite: 2310]. Which of the given options could be one of the two numbers [cite: 2310]?

A) 119 [cite: 2311]

B) 153 [cite: 2312]

C) 170 [cite: 2313]

D) 167 [cite: 2314]

SOLUTION EXPLANATION:

Let the two numbers be $17a$ and $17b$, where a and b are strictly co-prime numbers ($HCF(a, b) = 1$).

We know that: $LCM = HCF \times a \times b$.

Substitute our given values: $1309 = 17 \times a \times b \rightarrow a \times b = 1309 / 17 = 77$.

Now, find co-prime factor pairs that multiply to 77:

- Pair 1: $(1, 77) \rightarrow ext\{Numbers\} = (17 \times 1, 17 \times 77) = (17, 1309)$.
- Pair 2: $(7, 11) \rightarrow ext\{Numbers\} = (17 \times 7, 17 \times 11) = (119, 187)$.

Evaluating our options, the value 119 appears as Option A [cite: 2311].

Correct Answer: A [cite: 2311]

Question 4: Bezout's Identity Linear Combination Mechanics [cite: 2531]

If the Highest Common Factor (HCF) of 180 and 432 is expressed in the linear algebraic combination form $(180m + 432n)$, where m and n are integers [cite: 2531], determine the absolute difference between the coefficients m and n [cite: 2531].

A) 8 [cite: 2532]

B) 3 [cite: 2533]

C) 7 [cite: 2534]

D) 9 [cite: 2535]

SOLUTION EXPLANATION:

This targets Trap 2. First, let's find the HCF of 180 and 432 using the standard Euclidean division algorithm:

1. $432 = 180 \times 2 + 72 \rightarrow 72 = 432 - 180 \times 2$
2. $180 = 72 \times 2 + 36 \rightarrow 36 = 180 - 72 \times 2$
3. $72 = 36 \times 2 + 0$. Thus, $HCF = 36$.

Now, substitute the first equation into the second to express 36 as a linear combination:

$36 = 180 - 2 \times (432 - 180 \times 2)$

$36 = 180 - 2 \times 432 + 4 \times 180 = 180(5) + 432(-2)$.

Comparing this to our target expression $180m + 432n$, we get: $m = 5$ and $n = -2$.

The difference between m and n is: $m - n = 5 - (-2) = 5 + 2 = 7$ [cite: 2534].

Correct Answer: C [cite: 2534]

Question 5: Radical Floating-Point Decimal Operations [cite: 2492]

Determine the real evaluation value of the radical decimal expression given by: $(0.000314 + 0.000198)$^{(1/3)} [cite: 2492].

A) 0.8 [cite: 2493]

B) 0.04 [cite: 2494]

C) 0.08 [cite: 2495]

D) 0.4 [cite: 2496]

SOLUTION EXPLANATION:

This targets Trap 3. First, sum the floating-point values carefully:

$0.000314 + 0.000198 = 0.000512$.

Now, find the cube root of this decimal fraction: (0.000512)^{1/3}.

Convert the decimal to a clear fractional base to avoid mistakes:

$0.000512 = 512 / 1,000,000$.

Apply the cube root to both the numerator and denominator:

• Numerator: (512)^{1/3} $= 8$ (since 8 imes 8 imes $8 = 512$).

• Denominator: $(1,000,000)$^{1/3} $= 100$ (since 100 imes 100 imes $100 = 1,000,000$).

Thus, the result is: $8 / 100 = 0.08$ [cite: 2495].

Correct Answer: C [cite: 2495]

Question 6: HCF/LCM Common Modulo Remainder Theorem

Find the greatest 4-digit number that leaves remainders of 8, 11, 14, and 23 when divided by 12, 15, 18, and 27 respectively.

A) 9716

B) 9720

C) 9714

D) 9724

SOLUTION EXPLANATION:

Notice that the difference between each divisor and its corresponding remainder is constant:

$12 - 8 = 4;\ \ 15 - 11 = 4;\ \ 18 - 14 = 4;\ \ 27 - 23 = 4$.

This represents a fixed-difference negative remainder pattern. The required number can be expressed as:

$\text{LCM}(12, 15, 18, 27) \cdot k - 4$.

Let's find the LCM of our divisors:

- $12 = 2^2 \times 3$
- $15 = 3 \times 5$
- $18 = 2 \times 3^2$
- $27 = 3^3$

$\text{LCM} = 2^2 \times 3^3 \times 5 = 4 \times 27 \times 5 = 540$.

Now, find the largest 4-digit multiple of 540:

Dividing 9999 by 540: $9999 / 540 = 18.51$.

The highest 4-digit multiple is: $540 \times 18 = 9720$.

Subtract our fixed difference of 4 to get the final answer:

$9720 - 4 = 9716$.

Correct Answer: A

Question 7: Mixed Recurring Vinculum Notation Modification

Evaluate and simplify the sum of the mixed recurring decimals into a standard fraction: $0.2\overline{15} + 0.3\overline{42}$.

A) 551 / 990

B) 277 / 495

C) 553 / 990

D) 276 / 495

SOLUTION EXPLANATION:

Apply our recurring decimal shortcut formula to each term:

1. $0.2\overline{15} = (215 - 2) / 990 = 213 / 990$.
2. $0.3\overline{42} = (342 - 3) / 990 = 339 / 990$.

Now, add the two fractional values together:

$ext{Sum} = 213 / 990 + 339 / 990 = (213 + 339) / 990 = 552 / 990$.

Reduce the fraction to its lowest terms by dividing both the numerator and denominator by 2:

$552 / 2 = 276$ and $990 / 2 = 495$. This gives $276 / 495$.

We can simplify further by dividing by 3: $276 / 3 = 92$ and $495 / 3 = 165$.

Looking at our option choices, the expanded un-reduced form matches **276 / 495** directly.

Correct Answer: D

Question 8: Prime Prime-Factor Trailing Multipliers

Determine the total number of trailing zeros at the end of the calculated value of the factorial expression $125!$.

A) 25

B) 30

C) 31

D) 28

SOLUTION EXPLANATION:

The number of trailing zeros in a factorial depends on the highest power of 5 contained within that product sequence (since powers of 2 are always present in excess).

Apply Legendre's successive division shortcut theorem:

- Count 1: $125 / 5 = 25$
- Count 2: $25 / 5 = 5$
- Count 3: $5 / 5 = 1$

Sum all the quotients to find the total power of 5:

$ext{Total Trailing Zeros} = 25 + 5 + 1 = 31$.

Correct Answer: C

Question 9: Exponential Unit-Digit Cyclicity Scaling

Find the unit digit of the large product expression given by: $7^{295} \times 3^{158}$.

A) 1

B) 3

C) 7

D) 9

SOLUTION EXPLANATION:

Both base digits 7 and 3 follow a standard unit-digit pattern that repeats every 4 powers (cyclicity of 4). We can simplify the problem by looking at the remainders of the exponents when divided by 4.

- **For 7^{295}:** Divide 295 by 4. $295 = 73 \times 4 + 3$. The remainder is 3.

The unit digit matches $7^3 = 343 \rightarrow 3$.

- **For 3^{158}:** Divide 158 by 4. $158 = 39 \times 4 + 2$. The remainder is 2.

The unit digit matches $3^2 = 9$.

Now, multiply the two resulting unit digits together:

$ext\{Product Unit Digit\} = 3 \times 9 = 27 \rightarrow 7$.

Correct Answer: C

Question 10: Combinatorial Properties of Factors

Find the total number of even factors that exist for the composite integer 360.

A) 24

B) 18

C) 12

D) 6

SOLUTION EXPLANATION:

First, find the prime factorization of 360:

$360 = 36 \times 10 = 6^2 \times 2 \times 5 = (2 \times 3)^2 \times 2 \times 5 = 2^3 \times 3^2 \times 5^1$.

To find the number of *even* factors, the factor must contain at least one power of 2.

Shortcut: Do not add 1 to the exponent of prime base 2. Keep it as is, and add 1 to the exponents of all other prime bases normally:

$ext\{Even Factors\} = 3 \times (2 + 1) \times (1 + 1) = 3 \times 3 \times 2 = 18$.

Alternative Check: Total factors = $(3+1)(2+1)(1+1) = 4\ imes\ 3\ imes\ 2 = 24$. Odd factors (ignoring base 2) = $(2+1)(1+1) = 3\ imes\ 2 = 6$. Even factors = $24 - 6 = 18$.

Correct Answer: B